

Cagatay Ayhan

Applied Mathematics | Data Science | Machine Learning

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PROFESSIONAL SUMMARY

Biomathematics Ph.D. candidate who develops Python-based methods for extracting useful structure from complex scientific data. Experience spans topological data analysis, machine learning, model evaluation, visualization, and collaborative research with neuroscience and materials-science applications. Seeking data science, applied scientist, research scientist, and quantitative roles beginning Summer or Fall 2027.

TECHNICAL SKILLS

Programming and tools: Python, Git/GitHub, $\text{\LaTeX}$; NumPy, Pandas, SciPy, scikit-learn, Matplotlib, GeoPandas, PyTorch, Ripser, GUDHI

Data science and ML: exploratory data analysis, data cleaning, feature engineering, classification, ensemble models, model selection, cross-validation, imbalanced-data evaluation, dimensionality reduction, clustering, geospatial analysis

Applied mathematics: topological data analysis, persistent homology, optimal transport, statistical and computational modeling

RESEARCH EXPERIENCE

Research Intern

Summer 2026

Georgia Tech Research Institute

Atlanta, GA

- Investigated the use of topological data analysis to characterize structure in materials-science data under the supervision of Dr. Branden Stone.
- Translated an interdisciplinary research problem into a computational analysis workflow and communicated findings within a collaborative research setting.

Graduate Research Assistant

2024–2026

Florida State University, Department of Mathematics (NSF DMS-2324962)

Tallahassee, FL

- Developed an open-source **Python pipeline** that transforms multi-neuron spike-train recordings into persistent-homology features for classification and downstream analysis.
- Benchmarked the persistent-homology classifier against linear and RBF support vector machines across synthetic and biological datasets, identifying regimes where TDA better captures coordinated and higher-order population structure and where SVM performs better on rate-driven signals.
- Designed and implemented `bottleneck_zero`, a specialized near-linear-time algorithm for computing bottleneck distances between degree-0 persistence diagrams, translating theoretical results from **research on persistence-diagram distances** into efficient software used in the spike-train analysis pipeline.

SELECTED DATA SCIENCE & SOFTWARE PROJECTS

Visual TDA

Personal project

[GitHub repository](#) | [Interactive demo](#)

- Built interactive visual demonstrations of persistent homology to make filtrations, barcodes, and persistence diagrams easier to explore and communicate.
- Used the project as an interactive companion to the neural spike-train research paper, connecting mathematical exposition with reproducible visual software.

Decoding Human Activity

Summer 2025

Erdős Institute Data Science Boot Camp | [GitHub](#)

- Developed a classification workflow for human-activity recognition from time-series sensor data.
- Engineered predictive features and evaluated ensemble-learning approaches for multiclass activity classification in a collaborative project.

SELECTED DATA SCIENCE & SOFTWARE PROJECTS (CONTINUED)

Tallahassee Crime Map

Spring 2024

Erdős Institute Data Science Boot Camp | [GitHub](#)

- Analyzed 2023 public-safety data using spatial and temporal exploratory analysis to identify patterns and inform feature selection.
- Collaborated in a five-person team on a Pix2Pix conditional-GAN workflow that maps urban geometry to predicted crime heatmaps; evaluated predictions against a random baseline.

Near-Earth Object Classification

Fall 2023

Erdős Institute Data Science Boot Camp | [GitHub](#)

- Built a binary-classification pipeline for a dataset of more than 90,000 near-Earth-object observations, including exploratory analysis, preprocessing, and model comparison.
- Compared logistic regression, decision trees, random forests, and XGBoost; selected F1 score as the primary metric for the imbalanced target and achieved a peak F1 score of 96%.

SELECTED PUBLICATIONS

Cagatay Ayhan, Audrey N. Nash, Roberto Vincis, Martin Bauer, Richard Bertram, and Tom Needham. "A Persistent Homology Pipeline for the Analysis of Neural Spike Train Data." [arXiv:2512.08637](#), 2025. [Code]

Cagatay Ayhan and Tom Needham. "Equivalence of Landscape and Erosion Distances for Persistence Diagrams." [arXiv:2506.21488](#), 2025.

EDUCATION

Ph.D. Candidate in Biomathematics

Expected August 2027

Florida State University

Tallahassee, FL

M.S. in Biomathematics conferred Fall 2025 | GPA: 3.67/4.0

B.S. in Pure Mathematics

June 2021

Ege University

Izmir, Turkey

GPA: 3.70/4.0, High Honors

TEACHING & COMMUNICATION EXPERIENCE

Course Instructor and Teaching Assistant

2021–Present

Florida State University, Department of Mathematics

Tallahassee, FL

- Instructor of record for Applied Linear Algebra, Calculus I–III, Precalculus Algebra, and other undergraduate mathematics courses; currently teaching Applied Linear Algebra for the second time.
- Designed lessons, assessments, and explanations for students with varied technical backgrounds, strengthening stakeholder communication and presentation skills.
- Earned the Tam Family Award for Excellence in Teaching and Research and two nominations for the Outstanding Teaching Assistant Award.

LEADERSHIP & RECOGNITION

- **Co-organizer**, AMS Special Session on *Geometric Methods for Data Science*, 2026 Spring Southeastern Sectional Meeting.
- **First Place (tied)**, Dewitt Sumners Graduate Research Week Flash Talk Competition, 2025, for communicating topological data analysis to a broad audience.
- **Bettye Anne Bushee Case Graduate Fellowship**, 2024.